



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO GRG FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

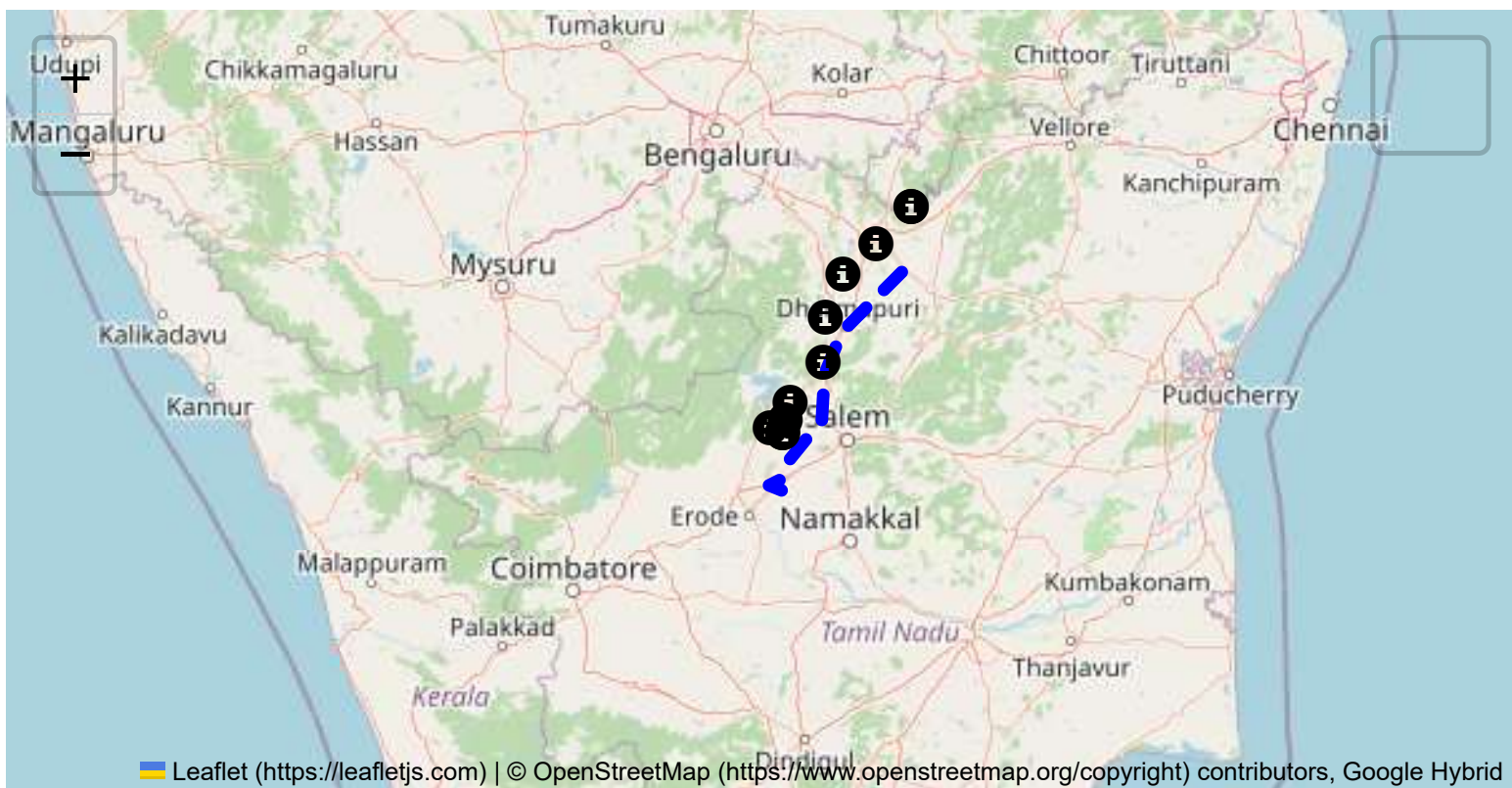
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

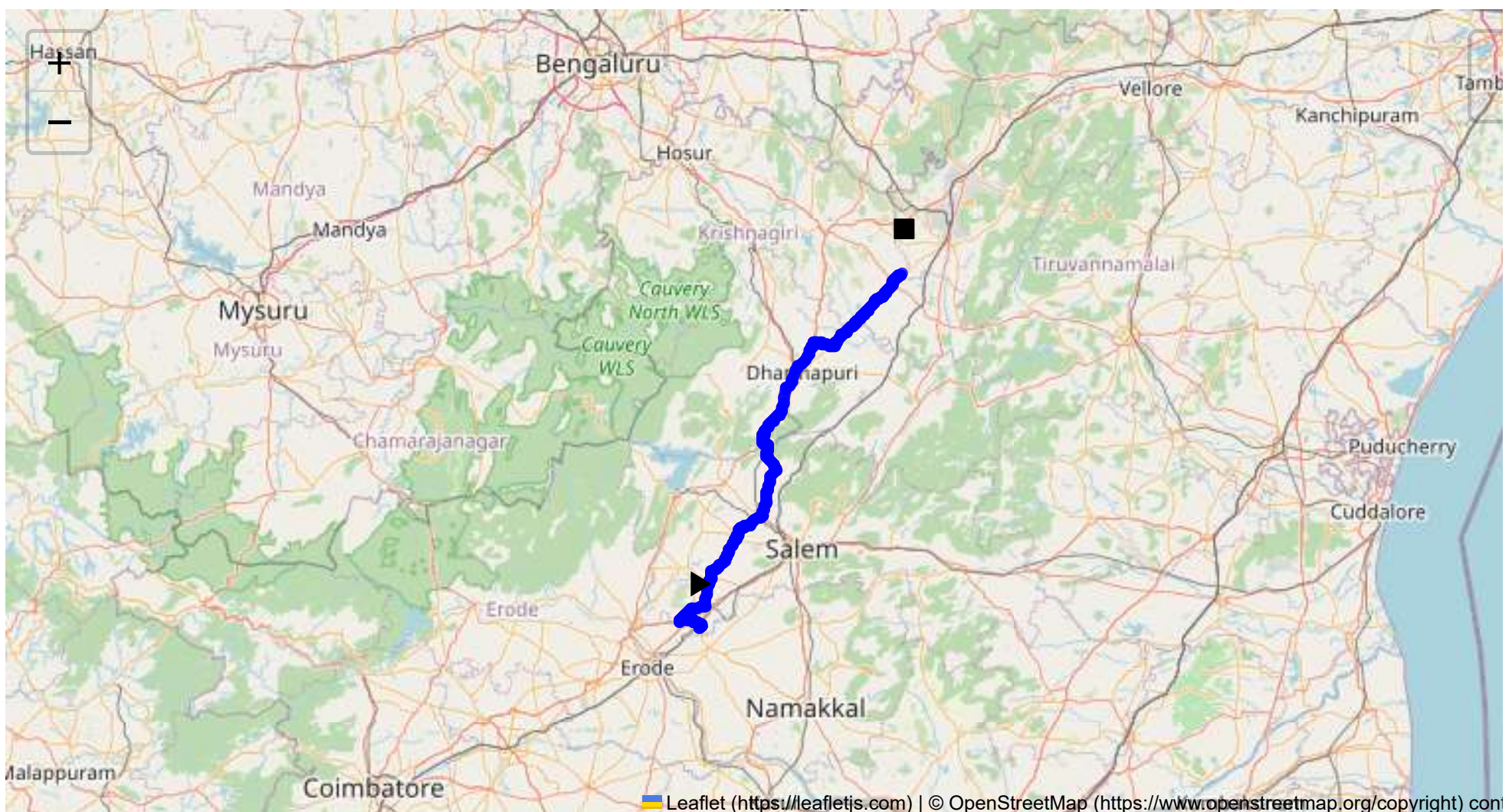
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 152.90 km
Estimated Duration: 3.3 hours
Adjusted Duration (Heavy Vehicle): 4.1 hours
Start: (11.4381, 77.8734)
End: (12.401855, 78.435288)



Welcome to the Journey Risk Management Study

Route Overview

The route from Sangagiri to Pudupatti traverses across a range of roads in Tamil Nadu, covering approximately 152.90 kilometers. The journey takes roughly 3.28 hours for heavy vehicles transporting hazardous materials and passes through key areas such as Puthur, Padaiveedu, and Omalur. The route mainly follows significant highways and state roads, including Salem - Kochi Highway and SH 86.

Weather Conditions and Hazards

Tamil Nadu generally experiences a tropical climate. Key weather-related hazards along the route include:

- **Monsoon Conditions:** Heavy rainfall from June to September and the northeast monsoon in October-November can lead to waterlogging and landslides, especially in hilly or mountainous areas.
- **Heat Waves:** During peak summer months (March to June), high temperatures can affect vehicle operations.

Traffic Patterns

Traffic peaks typically occur during morning and evening rush hours:

- Congestion-prone areas include intersections and stretches near major towns like Omalur and Thoppur.
- Market days or local festivals can add to traffic congestion unexpectedly.

Road Quality and Infrastructure

- **Road Conditions:** Major highways are generally well-maintained, but some state roads may have potholes or narrow sections.
- **Infrastructure:** Roadside amenities and signage are better on national highways compared to rural areas.

Alternative Routes for Emergencies

- **For Immediate Diversion:** Use of nearby state roads such as SH 86 when NH 44 sections near Thoppur are congested.
- Local roads often provide quicker access to facilities and bypasses in case of blockages or accidents.

Regulations Affecting Hazardous Materials

- Compliance with **Hazardous Materials Transport Regulations** in India is essential, requiring proper documentation and markings on heavy vehicles.
- Movement restrictions during high-traffic hours may apply in urban areas.

Historical Incidents

- There have been incidents involving heavy vehicles, especially around major highway intersections and curves. Precaution due to past accidents is advised at locations like Thoppur Ghat.

Environmental Considerations

- **Sensitive Areas:** Timberlands and water bodies around the route require careful consideration to prevent contamination or spills.

Communication Coverage

- **Coverage:** Generally robust along the highways, although some rural segments might experience weak mobile signals. \

Emergency Response Times

- Urban areas like Salem have rapid response capabilities (under 30 minutes), whereas rural locations might face delays due to distance and road conditions (up to an hour).

Overall Risk Assessment

- **Hazards:** Weather conditions, road quality, and traffic congestion can pose risks to heavy vehicles.
- **Mitigation:** Preparedness for monsoon conditions and alternative emergency routes can significantly reduce risk.

In conclusion, the proposed route has certain risks inherent to regions with varied geography and climate. Proper planning, adherence to local regulations, and readiness for weather and traffic-related challenges are crucial for the safe transport of hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43961, 77.87341	15 KM/Hr
1	Turn	High	11.44024, 77.87528	15 KM/Hr
2	Turn	High	11.44898, 77.87410	15 KM/Hr
3	Turn	High	11.45352, 77.85698	15 KM/Hr
4	Turn	Medium	11.45764, 77.85219	30 KM/Hr
5	Turn	High	11.46303, 77.84945	15 KM/Hr
6	Turn	Medium	11.45545, 77.81812	30 KM/Hr
7	Turn	High	11.45584, 77.81695	15 KM/Hr
8	Turn	Medium	11.45656, 77.81709	30 KM/Hr
9	Turn	High	11.49383, 77.88442	15 KM/Hr
10	Turn	Medium	11.56083, 77.89921	30 KM/Hr
11	Turn	Medium	11.56180, 77.90019	30 KM/Hr
12	Turn	Medium	11.59189, 77.90645	30 KM/Hr
13	Turn	Medium	11.59328, 77.90809	30 KM/Hr
14	Turn	Medium	11.63867, 77.94841	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
15	Turn	High	11.64217, 77.95200	15 KM/Hr
16	Turn	Medium	11.64712, 77.95118	30 KM/Hr
17	Turn	Medium	11.67858, 77.96834	30 KM/Hr
18	Turn	Medium	11.70062, 77.97961	30 KM/Hr
19	Turn	High	11.70218, 77.97866	15 KM/Hr
20	Turn	Medium	11.71399, 77.99580	30 KM/Hr
21	Turn	Medium	11.71674, 77.99901	30 KM/Hr
22	Turn	Medium	11.71917, 78.01123	30 KM/Hr
23	Turn	Medium	11.73769, 78.03451	30 KM/Hr
24	Turn	High	11.73651, 78.03871	15 KM/Hr
25	Turn	Medium	11.74143, 78.04613	30 KM/Hr
26	Turn	Medium	11.74265, 78.04710	30 KM/Hr
27	Turn	Medium	11.78005, 78.05788	30 KM/Hr
28	Turn	Medium	11.86928, 78.08384	30 KM/Hr
29	Turn	Medium	11.88009, 78.08164	30 KM/Hr
30	Turn	Medium	11.91277, 78.05853	30 KM/Hr
31	Turn	Medium	11.92667, 78.06517	30 KM/Hr
32	Turn	Medium	11.93213, 78.06468	30 KM/Hr
33	Turn	Medium	11.93480, 78.06255	30 KM/Hr
34	Turn	Medium	11.93746, 78.05331	30 KM/Hr
35	Turn	Medium	11.95531, 78.04698	30 KM/Hr
36	Turn	High	11.95854, 78.04990	15 KM/Hr
37	Turn	High	11.96015, 78.04945	15 KM/Hr
38	Turn	Medium	11.96266, 78.05115	30 KM/Hr
39	Turn	High	11.96433, 78.05113	15 KM/Hr
40	Turn	High	11.96542, 78.04938	15 KM/Hr
41	Turn	High	11.96650, 78.04927	15 KM/Hr
42	Turn	Medium	12.17027, 78.16812	30 KM/Hr
43	Turn	High	12.21894, 78.19193	15 KM/Hr
44	Turn	Medium	12.21623, 78.19627	30 KM/Hr
45	Turn	High	12.21372, 78.21163	15 KM/Hr
46	Turn	High	12.21458, 78.21217	15 KM/Hr
47	Turn	High	12.21205, 78.21630	15 KM/Hr
48	Turn	Medium	12.21231, 78.21741	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
49	Turn	High	12.21132, 78.22306	15 KM/Hr
50	Turn	Medium	12.21037, 78.22363	30 KM/Hr
51	Turn	High	12.20519, 78.24472	15 KM/Hr
52	Turn	Medium	12.20770, 78.24742	30 KM/Hr
53	Turn	Medium	12.26801, 78.30453	30 KM/Hr
54	Turn	Medium	12.26983, 78.30724	30 KM/Hr
55	Turn	Medium	12.28433, 78.32041	30 KM/Hr
56	Turn	Medium	12.34233, 78.38415	30 KM/Hr
57	Turn	Medium	12.34984, 78.38763	30 KM/Hr
58	Turn	Medium	12.37895, 78.41212	30 KM/Hr
59	Turn	Medium	12.38125, 78.41198	30 KM/Hr
60	Turn	Medium	12.40008, 78.42978	30 KM/Hr
61	Turn	Medium	12.40012, 78.43135	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
3	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
4	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
5	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
6	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
7	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
8	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
9	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
11	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
12	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
13	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
14	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
15	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
16	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
17	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
18	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
19	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
20	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
21	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
22	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
23	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
25	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
27	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
28	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
29	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
30	hospital	Government Hospital, Pochampalli	12.3322477, 78.365767	30 km/h	Medium
32	police	Mathur Police Station	12.3828409, 78.4125514	30 km/h	Medium
33	hospital	Government Hospital, Mathur	12.3826201, 78.4136738	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
10	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
24	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
26	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
31	school	GUNA Matric Higher Secondary School	12.3722333, 78.4086433	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44024, 77.87528



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44898, 77.87410



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45764, 77.85219



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45545, 77.81812



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45584, 77.81695



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45656, 77.81709



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49383, 77.88442



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56083, 77.89921



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.56180, 77.90019



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.59189, 77.90645



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.59328, 77.90809



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.63867, 77.94841



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64217, 77.95200



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64712, 77.95118



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67858, 77.96834



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.70062, 77.97961



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.70218, 77.97866



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71399, 77.99580



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71674, 77.99901



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71917, 78.01123



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.73769, 78.03451



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.73651, 78.03871



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.74143, 78.04613



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.74265, 78.04710



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.78005, 78.05788



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.86928, 78.08384



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.88009, 78.08164



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.91277, 78.05853



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.92667, 78.06517



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.93213, 78.06468



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93480, 78.06255



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93746, 78.05331



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95531, 78.04698



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.95854, 78.04990



Risk Type: Turn

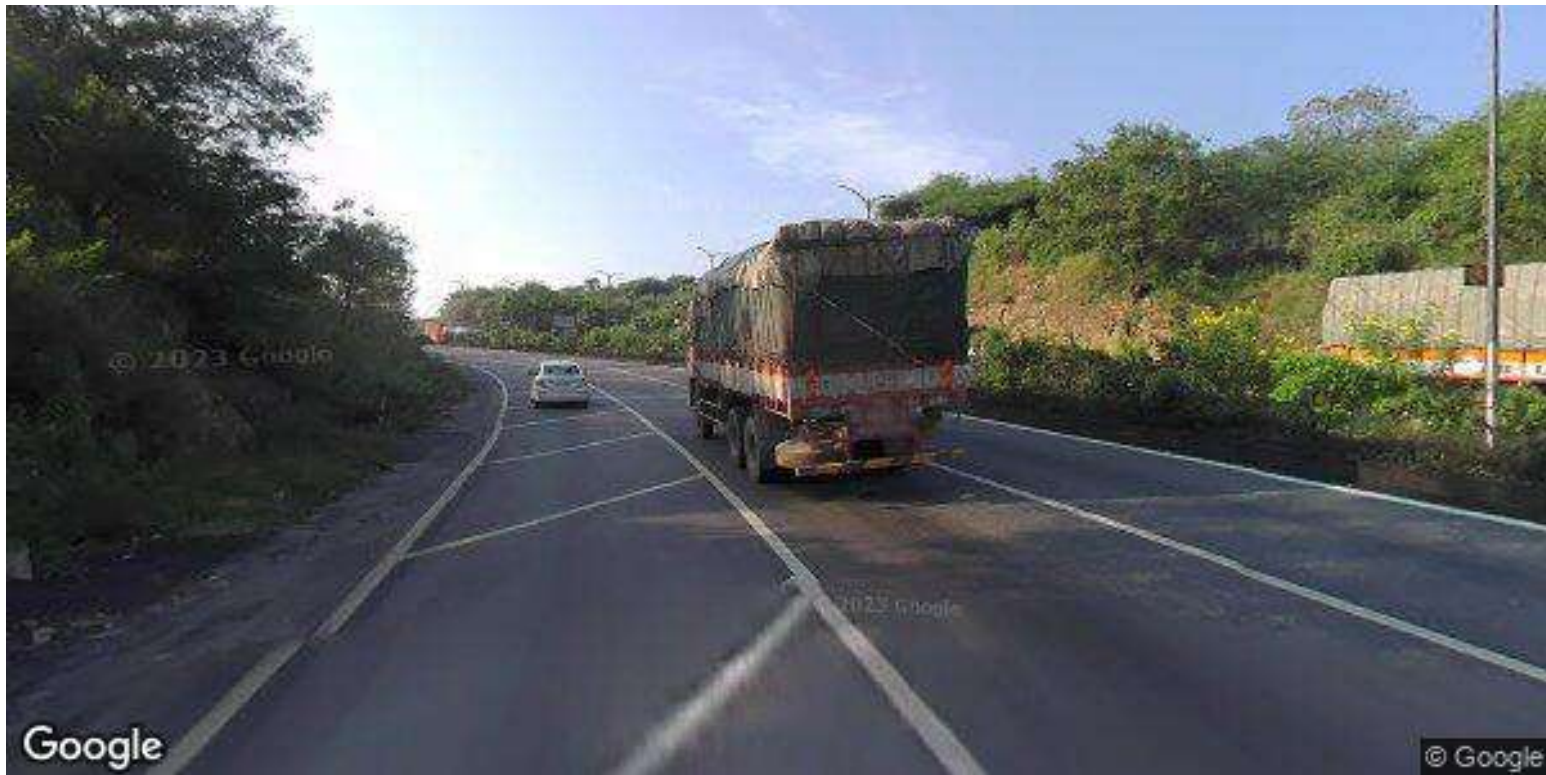
Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96015, 78.04945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96266, 78.05115



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96433, 78.05113



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96542, 78.04938



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96650, 78.04927



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.17027, 78.16812



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.21894, 78.19193



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.21623, 78.19627



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.21372, 78.21163



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.21458, 78.21217



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.21205, 78.21630



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.21132, 78.22306



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.21037, 78.22363



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.20519, 78.24472



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.20770, 78.24742



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.26801, 78.30453



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.26983, 78.30724



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.28433, 78.32041



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.34233, 78.38415



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.34984, 78.38763



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.37895, 78.41212



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.38125, 78.41198



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.40008, 78.42978



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.40012, 78.43135
